

Non-Destructive Reverse Engineering of a Closed Android–ROS Service Robot: Open ROSBRIDGE Integration and Edge Conversational Stack Without Vendor Support

Claudia LUSAMOTE KIMFUTA
HESTIM Engineering & Business School
Email: clusamote@gmail.com

Dr. Sridath Tula
HESTIM Engineering & Business School
Supervisor

Abstract—Commercial service robots often ship with proprietary Android applications and undisclosed communication protocols, preventing customization of reception flows, guided tours, and conversational assistants. We present CYBEL, an open edge platform built by *non-destructive reverse engineering* of the CIOT TY1251D-03195 reception robot a dual-processor system combining a ROS navigation chassis and an Android 7.1 head without manufacturer documentation or support. Our seven-phase methodology combines network mapping, ROSBRIDGE/roapi introspection, static APK analysis (JADX) [1], passive MQTT observation, and field validation using rule H4: a successful ROSBRIDGE transport acknowledgment does not guarantee motor execution. We document reconstructed teleoperation (/cmd_vel_mux/input/teleop), autonomous navigation (/navi_goal vs. POI chain /tag_manager/navi), and an edge conversational layer local knowledge engine, Android TTS bridge, guided tour orchestration replacing the vendor welcome application. A hybrid strategy reusing POIs from the manufacturer Deployment Tool (Sentrymove) improves navigation reliability over coordinate-only goals. Field results show <100 ms local API latency, ~90% ROSBRIDGE session reliability, and an autonomous visitor kiosk on Termux.

Index Terms—Service robotics, reverse engineering, ROS, ROSBRIDGE, embedded Android, human-robot interaction, edge computing, text-to-speech

I. INTRODUCTION

Mobile reception robots integrate mature SLAM and motion planning with touchscreen and speech interfaces, yet remain software black boxes: no public SDK, no message schemas, and often mandatory vendor cloud services. For research labs and training institutions, this closure blocks custom visit paths, domain-specific knowledge bases, and open conversational agents.

We address the question: *How can a third-party command and interaction platform match or exceed a proprietary Android-ROS welcome application when no official interface documentation exists?* Sub-problems include discovering network entry points, reconstructing message formats, validating causal command effects, rebuilding speech when ROS exposes no TTS channel, and operating on closed robot Wi-Fi without guaranteed Internet.

Contributions (project CYBEL, HESTIM 2026):

- C1. A reproducible seven-phase non-destructive reverse-engineering pipeline for closed vendor robots.
- C2. Documented ROSBRIDGE v2 reconstruction for teleop, POI/coordinate navigation, and telemetry.
- C3. Edge-first architecture: interchangeable MockRobot/RealRobot SDK, dual PC/Termux backend, autonomous WebView kiosk.
- C4. Empirical TTS resolution via a minimal Android broadcast bridge when ROS, HTTP, and MQTT fail.
- C5. Open conversational layer (local FAQ + robot actions), extensible to LLMs without vendor cloud.
- C6. Hybrid Sentrymove POI reuse strategy improving field navigation over raw coordinates.

Scope. This paper focuses on protocol discovery and open-stack integration; LLM chatbots are architected but not yet deployed. The codebase and exploration scripts are maintained as reproducible artifacts.

II. ROBOT PLATFORM AND NETWORK TOPOLOGY

A. Dual-processor hardware

The CIOT TY1251D-03195 combines:

- **Chassis:** Linux + ROS (Noetic/Melodic), SLAM, move_base, LiDAR/RGBD, ROSBRIDGE, MQTT broker.
- **Head:** Android 7.1 (RK3399, 2 GB RAM) - 15.6" touchscreen, cameras, speakers, vendor apps.

CYBEL never obtains shell access on the chassis (SSH locked; brute-force triggers lockout). Usable channels: WebSocket ROSBRIDGE on port 9090 and MQTT on 1883 (observation only).

B. Triple network segment

Mapping revealed three distinct segments (Fig. 1):

- **Robot AP** (TY1251D-03195): chassis at 10.42.0.1:9090.
- **Internal eth0:** head 192.168.20.1 ↔ chassis 192.168.20.22:9090, preferred path from Termux.
- **Lab Wi-Fi:** head 172.16.0.x (DHCP); PC uses ADB for deployment.

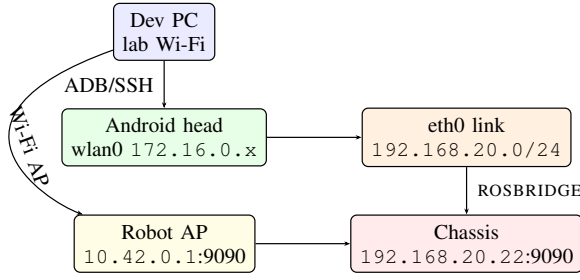


Fig. 1. Network topology. Termux on the tablet uses 192.168.20.22; confusing head DHCP IP with chassis IP causes silent failures.

TABLE I
REVERSE-ENGINEERING PHASES AND DELIVERABLES

#	Phase	Key deliverable
1	Network scan	Dual-IP map, ports 9090/1883
2	ROS observation	Topic/service inventory
3	MQTT passive	No movement control via MQTT
4	APK audit (JADX)	Topic parity vs. Sentrymove
5	Wireshark	WebSocket JSON framing
6	ADB / Termux	TTS bridge, edge backend
7	Field validation	phase0_robot_check.py

Vendor packages identified via ADB: com.ciot.welcomepatrol (welcome UI), com.ciot.sentrymove (Deployment Tool), shared library selfchassislibrary treated as *de facto* protocol specification after JADX audit.

III. REVERSE ENGINEERING METHODOLOGY

A. Guiding principles

- P1.** Observe before publishing (passive MQTT/ROSBRIDGE first).
- P2.** Verify real effects rule **H4**: after teleop publish, /robot_pose must change.
- P3.** Triangulate network traces, rosapi, decompiled APK, and field logs.
- P4.** Remain non-destructive: no firmware flash, no chassis root.
- P5.** Mock-first development before RealRobot integration.

B. Seven-phase pipeline

Table I summarizes the pipeline; each hypothesis follows *observe* → *isolated script* → *telemetry check* → *SDK integration or documented rejection*.

Rejected hypotheses (documented): legacy teleop topic /mobile_base/commands/velocity; candidate TTS topics (/play_tts, /robot_tts); HTTP TTS on head ports 80/8080/8888; MQTT movement commands.

IV. ROSBRIDGE PROTOCOL RECONSTRUCTION

A. Transport and client

ROSBRIDGE v2 over WebSocket TCP 9090 [2], [3]; JSON operations subscribe, publish, advertise, call_service. Our RosbridgeClient (sdk/rosbridge.py) uses three connection retries (20s timeout), 20s ping keepalive, and automatic reconnect.

TABLE II
NAVIGATION MODES AND PREREQUISITES

Mode	Mechanism	Gate
Coordinates	/navi_goal	loc. $\geq 60\%$, status 601
Named POI	/tag_manager/navi → /poi	POI in map

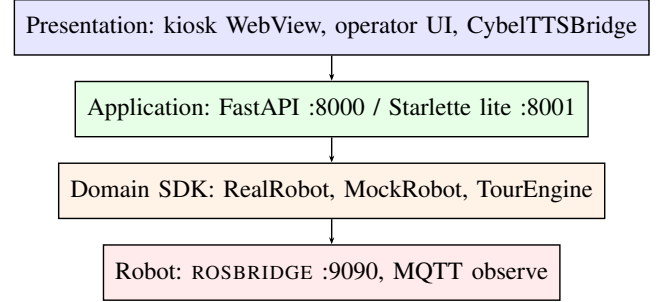


Fig. 2. Logical architecture (edge-first, no vendor cloud).

B. Teleoperation

Manufacturer-confirmed topic:

```
/cmd_vel_mux/input/teleop
(geometry_msgs/Twist)
```

Sequence: advertise then periodic publish (~ 10 Hz). Sentrymove ramps linear velocity $\pm 0.025/100$ ms with $\omega_z \times 0.8$ scaling.

Gap E1: early CYBEL builds blocked teleop unless manual mode was active (control_state==30); Sentrymove does not require this step. **Fix**: auto-enable manual mode on first move().

C. Autonomous navigation

Two modes coexist on the chassis:

nav_status codes gate CYBEL commands: 600 not localized, 601 ready, 602 navigating, 603 arrived, 604 error. Cancellation chain: /move_base/cancel, /path_follower/cancel, /poi{command:stop}, zero velocity on mux.

D. Telemetry throttling

Subscriptions use throttle_rate (ms): /robot_pose 200 (~ 5 Hz), /navi_status 500, LiDAR /scan_filter 100 — limiting ROSBRIDGE saturation on constrained Wi-Fi.

V. CYBEL SOFTWARE ARCHITECTURE

A. Three-layer stack

Presentation: frontend-kiosk/ visitor UI; CybelVisitorKioskTest APK wraps WebView; CybelTTSBridge handles speech.

Application: FastAPI on PC (port 8000) for development; Starlette cybel_lite.py on Termux (port 8001 for POI variant) for standalone kiosk without PC.

Domain: RobotBackend protocol with MockRobot/RealRobot implementations — 83 unit tests pass without physical robot.

TABLE III
TTS INVESTIGATION SUMMARY

Channel tested	Result
ROS publish topics	No subscribers
ROS services	Absent in <code>rosapi</code>
HTTP head (80, 8080, 8888)	Closed
MQTT	Odometry only
ADB broadcast	Working (CybelTTSBridge)

B. Edge deployment pattern

Termux on the tablet runs Python, serves the kiosk over `0.0.0.0:8001`, connects to `192.168.20.22:9090`, and triggers TTS via local am broadcast. The visitor session requires no PC after deployment (`deploy_termux.py` or ADB bundle).

VI. CONVERSATIONAL LAYER WITHOUT VENDOR SUPPORT

The vendor `welcomepatrol` app couples Iflytek TTS via opaque IPC and a proprietary cloud knowledge API. CYBEL replaces this with an *open edge conversational pipeline*:

- 1) **Input:** kiosk touch, operator text, or browser Web Speech API (PC only).
- 2) **KnowledgeEngine:** keyword scoring over local JSON (`knowledgeV2-labV2.json`, FAQ) — threshold ≥ 2.0 ; deterministic, offline-capable.
- 3) **ReceptionService:** `matched answer` → `RobotSpeech.speak()` + `optional navigate_to_point()`.
- 4) **TourEngine:** ordered stops from `lab_tour.json` with speech/navigation ordering fix (relocalize *before* TTS to avoid “speaks but does not move”).

A. TTS channel discovery

No ROS topic, service, HTTP port, or MQTT payload exposed TTS on TY1251D (Table III) [4].

CybelTTSBridge (~50 KB APK): `BroadcastReceiver` → Android `TextToSpeech` (Google engine, French locale):

```
am broadcast -n com.cybel.ttsbridge/.SpeakReceiver
-a com.cybel.ttsbridge.SPEAK --es text "Welcome."
```

`RobotSpeech` tries ROS, HTTP, ADB-from-PC, then local Termux broadcast — documented fallback chain in `sdk/speech.py`.

LLM extensibility: `POST /api/knowledge/ask` accepts the same contract; an LLM backend can call validated CYBEL navigation/speech APIs without rediscovering ROS.

VII. HYBRID NAVIGATION CASE STUDY

A. Coordinate-only failure (strategy S1)

Publishing `/navi_goal` with coordinates from a JSON knowledge base often failed in the field: robot spoke but `nav_status` remained 601 (planner never entered 602), wrong destinations due to map drift, slow relocalization.

Tour trace `tour_20260623_142601`: goal published, no 601→602 transition.

TABLE IV
OBSERVED PERFORMANCE INDICATORS

Metric	Value
Local REST latency	<100 ms
ROSBIDGE connect	2–60 s (3×20 s retry)
Session reliability	~90% (Wi-Fi dependent)
Wi-Fi RTT	89–1654 ms
Unit tests	83/83 passed
Coord. navigation reliability	Low (motivates POI pivot)
Termux kiosk autonomously	Validated

B. Sentrymove POI reuse (strategy S3)

Sentrymove (Deployment Tool) navigates reliably via named POIs on map `laboV2`. CYBEL hybrid flow:

- 1) Create POIs in Deployment Tool (MAJUSCULES-TIRETS naming).
- 2) Sync ROS markers → `data/points.json` (`sdk/poi_sync.py`).
- 3) Kiosk auto-sync on destination grid load; tour uses `target_point` in `lab_tour.json`.
- 4) A/B test: port 8000 (coordinates) vs. 8001 (POI) on same hardware.

Gap analysis [5] drove fixes: mux teleop topic, POI service chain in lite backend, localization threshold 60%, cancellation parity.

VIII. EVALUATION AND DISCUSSION

A. Metrics (June 2026, lab robot)

Validation script: `phase0_robot_check.py`
`-host 192.168.20.22 -teleop -nav-poi "CNC ROUTEUR" -tts.`

B. Lessons for closed robot integration

- **Transport lies:** ROSBRIDGE accepts publishes without motor effect; telemetry is ground truth.
- **Vendor APK as spec:** JADX of Sentrymove accelerated discovery vs. generic ROS docs.
- **Dual-IP is normal:** document head vs. chassis addresses explicitly.
- **Reuse over reimplement:** hybrid POI avoids replicating `marker_manager`.
- **Edge first:** closed robot Wi-Fi precludes cloud LLM/TTS dependencies.

C. Limitations

Chassis shell inaccessible; no native ROS TTS; Android 7.1 WebView constraints; DHCP head IP variability; on-robot ASR not yet implemented; ethical APK decompilation for interoperability only.

IX. CONCLUSION

We presented non-destructive reverse engineering of the CIOT TY1251D-03195 and the open CYBEL platform: seven-phase methodology, validated ROSBRIDGE reconstruction, edge architecture with mock/real SDK, Android TTS bridge, local conversational layer, and hybrid Sentrymove POI navigation.

Future work: LLM/RAG behind `/api/knowledge/ask`, on-device ASR, presence-triggered kiosk wake-up (`/detected_people_array`), and publication of exploration scripts as reproducible artifacts for similar closed Android-ROS robots.

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